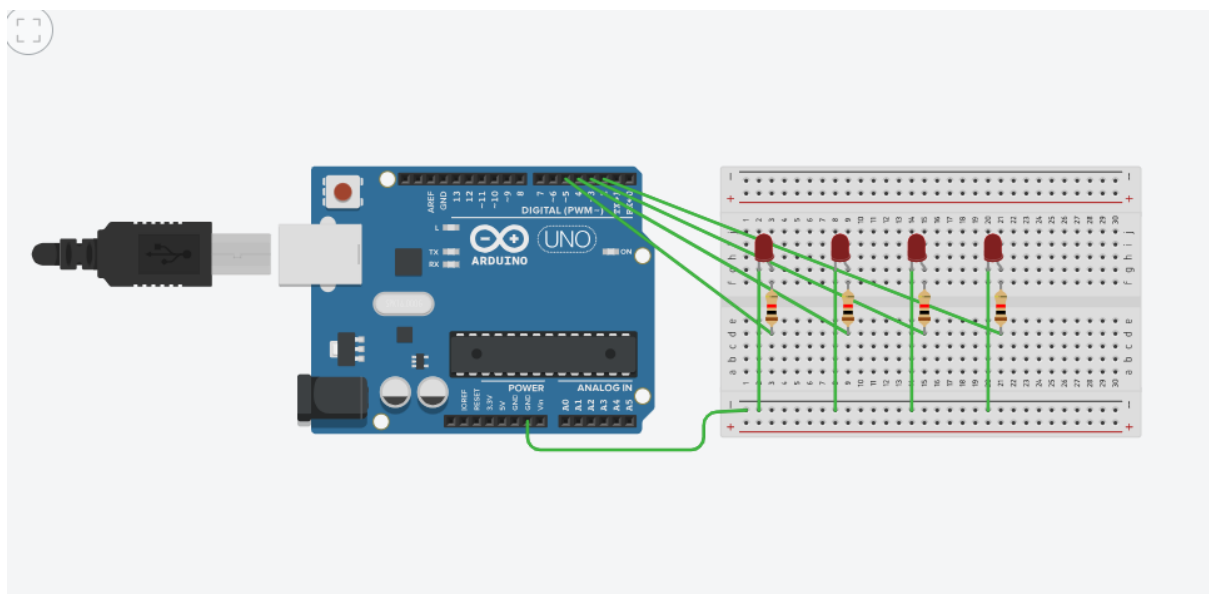


Robocon- Tinkercad Task 01

Circuit diagram and code explanation

Task- Decimal to Binary Conversion Using LEDs in Tinkercad

Circuit:



Explanation:

Components used— Arduino UNO, 4 LEDs, 4 Resistors, Breadboard, jumper wires

Code logic—

- First read the number and store it in a variable (input) using `Serial.parseInt()`
- Then check whether the input is in valid range i.e from 0 to 15

- For numbers lying within the range we use some operations to convert from decimal to binary
 - 1st store the input value temporarily in another variable(temp)
 - Use $\text{temp} \% 2$ which would give one of the two output 0 or 1 which would correspondingly refer to off and onn on LED
 - To carry on the loop, to find all 4 value combination of the decimal number converted to binary we use $\text{temp} = \text{temp} / 2$
- To turn on and off the corresponding led we use an array to arrange the led output sequence for the loop